

Kingston Khai

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Education

Aptech, Advanced Diploma in Software Engineering (ADSE)

Skills

Languages: Python, TypeScript, JavaScript, Java, Golang, Dart, Swift

Frameworks: FastAPI, Django, Spring Boot, NestJS, Next.js, Flutter, Gin

Databases: PostgreSQL, MongoDB, MySQL

Tools: Git, Postman, Figma, Expo, Android Studio, XCode, Docker

Experience

HNG Tech, Software Engineering Intern

Oct 2025 – Dec 2025

- Developed high-throughput RESTful APIs with FastAPI/Django using inverted index search and TF-IDF ranking for semantic NLP queries (sub-200ms latency at p95)
- Contributed to Legal WatchDog's AI compliance engine using graph traversal to map regulatory dependencies, reducing manual compliance efforts by 50% and accelerating audit cycles by 3×
- Built a document analyzer using OpenRouter's LLM APIs with chunking and embedding similarity search, achieving 92% summarization accuracy

SnapILabs, Frontend Developer Intern

Aug 2024 – Nov 2024

- Built SupaNote's AI workspace landing page with virtualized scrolling and debounced search, achieving 98+ Lighthouse score and 40% lower bounce rate
- Rebuilt Krea AI's UI from static mockups using component composition, delivering pixel perfect responsive implementation 2 weeks ahead of schedule
- Implemented a mobile-first design system in Tailwind CSS with atomic CSS architecture, accelerating feature development velocity by 30%

Projects

FarmIntel, AI-Powered Agricultural Intelligence Platform

React, Python, Django, PostgreSQL, Hugging Face Transformers

- Engineered plant disease detection pipeline using a fine-tuned Vision Transformer (ViT) trained on 15K+ annotated images with 98% accuracy and <1.2s inference latency
- Architected RBAC for 4 entity types using Django middleware, JWT sessions, and OTP/Google OAuth, reducing unauthorized access by 92%
- Integrated Paystack billing with idempotent webhook handling and token bucket rate limiting for B2B clients

Hirecraft, Domestic Services Marketplace

React, Java, Spring Boot, PostgreSQL

- Built two-sided marketplace with Spring Boot REST APIs featuring R-tree spatial indexing for location-aware provider search within 5km, boosting match success rate by 45%
- Implemented booking state machine with idempotent transaction handling, reducing scheduling conflicts by 88% during peak demand
- Secured payment flow via Paystack with PCI-DSS compliant tokenization and webhook signature verification, achieving zero fraud incidents

Leadership

HNG Tech, Engineering Team Lead

- Spearheaded a 3-engineer pod to architect and deploy a distributed notification microservice using Redis Streams pub/sub with idempotent delivery, reducing notification latency by 40% (p95 < 300ms)
- Championed CI/CD adoption (GitHub Actions + Docker) that accelerated deployment frequency by 3× and eliminated staging environment drift

Aptech, Lead Hackathon Developer

- Spearheaded end-to-end UI redesign of a pet health mobile app serving veterinary appointment booking, medication tracking, and telehealth features; ensuring a scalable design system with reusable flutter widgets
- Architected a scalable Flutter design system with BLoC state management and lazy loading optimizations, cutting cold-start time by 42% (2.8s → 1.6s)
- Built an interactive browser-based number guessing game with Flask and Jinja2 templates with the inclusion of a real time leaderboard across 3 difficulty levels

Certifications

- Udemy, Make a 2D Action Platformer in Unreal Engine 5
- NVIDIA, Building RAG Agents with LLMs